TRANSIENT THERMAL ANALYSIS OF ACTIVE AND PASSIVE INTERVENTIONS FOR NEXT-GENERATION WILDLAND FIREFIGHTER EMERGENCY SHELTERS

Ana Gegetchkori[1], Francisca Gomez Weiss[1], Kristian Nasi[1], Jaylen Patel[1],

Chris Schaefer[1], Ayden Wells[1]

[1]Department of Aerospace and Mechanical Engineering, Saint Louis University

St. Louis, MO, USA

This paper presents a preliminary analysis of two emergency fire shelter interventions designed to improve thermal protection while maintaining practical limits on weight, volume, cost, and deployment. Wildfires are increasing in frequency and intensity, with approximately 112.3 million hectares burned globally in 2026 and U.S. firefighter entrapment incidents reaching a 20-year high. The 2013 Yarnell Hill Fire, which claimed 19 firefighters despite shelter deployment, highlights the need for improved thermal protection.

During critical burn-over events, emergency fire shelters serve as the last line of defense against extreme temperatures reaching 2192°F (1200°C) and radiant heat fluxes exceeding 27 kW/m². Conventional shelters reflect radiant heat through an aluminum foil-laminate exterior but provide limited thermal buffering once heat penetrates the barrier. Internal heat accumulation, metabolic heat, and elevated CO_2 concentrations can further compromise survivability during prolonged entrapment. Previous improvements have primarily focused on insulation and weight reduction, leaving alternative cooling strategies relatively unexplored.

This study evaluates two interventions: a passive Phase Change Material (PCM) layer and an active CO_2 heat exchanger. Using a transient lumped-parameter thermal model, both designs were evaluated against the standard M-2002 shelter, which reached an internal temperature threshold of 122°F (50°C) after 11.8 seconds under the modeled exposure. The Paraffin RT42 PCM layer extended this time to 63.1 seconds but added 4.66 kg, while the single-pass CO_2 manifold extended it to 39.1 seconds, achieving a peak heat removal rate of 1.10 kW. These findings highlight the trade-offs between thermal performance, cooling capacity, and portability, providing direction for future shelter optimization.